Day 9: Linux System Administration Notes

Phase 2: Linux & Computer Fundamentals | Focus: Processes, Services, Users & Env Vars

1. Processes & Execution (ps, top, kill)

* Process ID (PID): Unique numerical ID assigned to every active process loaded into RAM.
* PID 1 (systemd / init): The ancestor of all processes started by the Kernel during boot.
* ps aux: Snapshot of all running processes across all system users.
* top / htop: Real-time interactive view of CPU, RAM usage, and running process threads.
* Graceful vs Forceful Termination:
  - kill <PID> (SIGTERM / Signal 15): Requests graceful shutdown (flushes files & closes DBs).
  - kill -9 <PID> (SIGKILL / Signal 9): Instantly terminates process at Kernel level without cleanup.

2. Background Services & Daemons (systemctl)

* Daemons: Background processes (e.g. sshd, dockerd) running continuously without a terminal.
* Service Management Commands (systemd):
  - sudo systemctl status <service>: Inspect status, logs, and uptime.
  - sudo systemctl start/stop/restart <service>: Control active execution state.
  - sudo systemctl enable <service>: Configure service to auto-start on machine reboot.

3. Users, Groups & Privilege Escalation (sudo)

* Standard User vs Root (uid=0): Standard users are restricted to their home directories (~/). Root has unrestricted system power.
* Privilege Escalation (sudo): Allows authorized users to execute administrative tasks with root privileges securely without logging in permanently as root.

4. Environment Variables & Persistence (~/.bashrc)

* Environment Variables: Dynamic key-value pairs stored in RAM (export KEY='value').
* Session Scope: An exported variable exists only inside the current terminal process.
* Persistence (~/.bashrc): To persist variables across sessions, append exports to ~/.bashrc and apply changes immediately using: source ~/.bashrc
* $PATH Variable: Colon-separated list of directories Linux scans to locate executable binaries.
* Package Management (apt):
  - sudo apt update: Updates local index of available software packages.
  - sudo apt install -y <pkg>: Installs system software directly from official repositories.